

Derivational Affixes as Heads and Phrases: Evidence from Turkish Diminutives

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Abstract

Recently the Distributed Morphology literature hosted two strands of debates regarding the status of derivational affixes. One main strand has been the root versus functional head status of derivational affixes. While [Marantz \(1997\)](#) and [Marvin \(2002\)](#) (among others) treated derivational affixes as functional heads, [Lowenstamm \(2014\)](#) and [De Belder \(2011\)](#) argued that they are acategorical roots. On the other hand, [Creemers, Don, and Fenger \(2018\)](#) argued that some affixes are roots and others are functional heads. The second strand of discussion has been the head versus modifier status of affixes in general. [Wiltschko and Steriopolo \(2007\)](#) showed that some affixes are heads whereas others are modifiers, and [Gouskova and Bobaljik \(2022\)](#) proposed that some affixes can be both. In this paper, we analyze the diminutive suffix *-Cik* in Turkish and argue that it can attach to its base as a head or as a phrasal modifier by being embedded in a larger phrase. Our proposal accounts for a variety of syntactic, semantic, and morphophonological properties such as variable productivity, vowel epenthesis, stress shift, and special idiomatic interpretations associated with *-Cik*.

1 Introduction

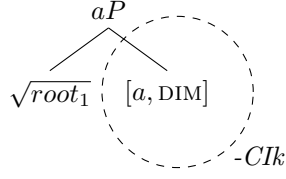
The syntactic structure of derivational affixes has been a matter of debate in the recent Distributed Morphology literature. Early accounts diverged on whether derivational affixes should be analyzed as functional heads or as roots. Notably, [Marantz](#)

(1997, 2007) and Marvin (2002), among others, treated derivational affixes as category-defining functional heads that combine with a root or an already categorized phrase. In contrast, Lowenstamm (2014) and De Belder (2011) argued that they are acategorical roots that directly combine with other roots. More recently, accounts that go beyond the simple dichotomy have been proposed. Creemers et al. (2018) took a middle ground arguing that some affixes are roots whereas others are category-defining functional heads. In a related but different line of research, Wiltschko and Steriopo (2007) showed that some affixes are heads whereas others are modifiers. Likewise, Gouskova and Bobaljik (2022) proposed that sometimes a single affix can function as a head or as a modifier. They also argued that some affixes possess internal complexity, exhibiting properties characteristic of phrasal structure.

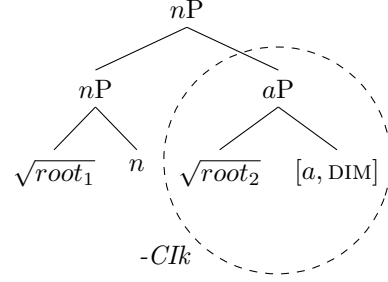
- (1) a. küçük – küçü-cük
small – small-DIM
'small – very small' (adjectival diminutive)
- b. kedi – kedi-cik
cat – cat-DIM
'cat – little cat' (nominal diminutive)

¹Throughout the paper, we use the terms ‘adjectival root’ and ‘nominal root’ a bit loosely. We do not assume that roots carry category information. Instead, what we intend to convey with this descriptive terminology is some root in a diminutivized word, where the word is either an adjective or a noun. For example, we call the root in an adjectival diminutive (e.g., *küçücük* ‘very small’) an adjectival root.

(2) a. adjectival diminutive



b. nominal diminutive



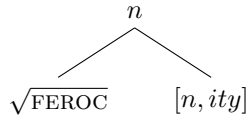
The organization of the paper is as follows. Section 2 provides a brief overview of the literature discussing the status of affixes as roots, functional heads, and internally complex modifiers to establish the tests we use in analyzing the nature of the diminutive suffix in Turkish. Section 3 presents the properties of the adjectival and nominal diminutives in Turkish. Section 4 first lays out the core assumptions of our proposal and builds a morphosyntactic and a compositional semantic account of the Turkish facts. Section 5 concludes the discussion by tying up some loose ends and providing a summary.

2 Background

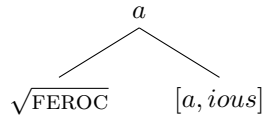
Derivational affixes have long been considered category-defining heads (Halle & Marantz, 1993; Selkirk, 1982; Williams, 1981). Marantz (1997, 2007) proposed the idea that lexical categories do not project from the lexicon but are instead projected from category-defining functional heads that combine with lexical roots. These category-defining functional heads are considered to be phase heads that trigger cyclic spell-out of the derivation, accounting for the cyclic phonological effects (Chomsky & Halle, 1968; Kiparsky, 1982). A simplified illustration of this idea is given in (3) with examples from Embick and Marantz (2008) and Murphy (2024), where the ultimate form and category of a root is determined by the category-defining head.

(3) category-defining functional heads

a) *ferocity*

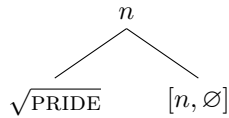


b) *ferocious*

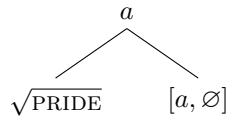


(Embick & Marantz, 2008, 23)

c) *pride*



d) *proud*



(Murphy, 2024)

The idea that category-defining heads are phasal (Arad, 2003; Embick, 2010; Embick & Marantz, 2008; Marantz, 2007; Marvin, 2002) has a range of properties that allow accounting for cyclic phonological effects, such as distinguishing stress-shifting and stress-neutral affixes as well as idiosyncratic interpretations and selectional restrictions. Embick and Marantz (2008) draw a distinction between the *inner domain* (root + first category-defining head) and the *outer domain* (category-changing heads). Morphemes² attaching in the inner domain may lead to idiosyncratic interpretations as well as special allomorphy whereas morphemes in the outer domain cannot due to locality (i.e. being in different phases).³

The idea that derivational affixes are category-defining functional heads has not gone unchallenged. Lowenstamm (2014) criticized the idea on two empirical grounds: 1) stress-shifting properties of certain affixes cannot be accounted for in such a theory and 2) some affixes are flexible in their categorial nature. For stress-shifting properties, Lowenstamm (2014) showed that a phase-based category-defining functional head view makes wrong predictions regarding stress-shifting affixes implementing two different spell-out views: i) phase heads spell out their complements (Marvin, 2002) and ii) phase heads spell out cyclic/phasal complements (Embick, 2010). The predictions of both views are shown in Table 1 for the English word *atomicity* where stress shifts to the penultimate syllable with the addition of each suffix. The assumption is that stress is assigned at the spell-out and the stress assigned in the first spell-out would not be accessible at later spell-outs.

Table 1 Predictions of *derivational morphemes as functional heads* in a phasal syntax.

English pattern →	<i>átom</i>	<i>atómic</i>	<i>atomicity</i>
Marvin (2002) spell-out	[átom]	*[átom]ic	*[átom]icity
Embick (2010) spell-out	[átom]	[atómic]	*[atómic]ity

(‘) indicates stress, brackets ([]) indicate the spell-out domain, asterisk (*) indicates unacceptability (wrong form). Examples adapted from Lowenstamm (2014, 10).

The second challenge, presented first by De Belder (2011) and independently by Lowenstamm (2014), is the *flexibility* of certain affixes in terms of what they “output” after they combine with a root (or a larger base). Some affixes can combine with different roots (or bases) and the resulting category can be more than one. For example, the English suffix *-ian* can return a noun or an adjective depending on what it combines with as shown in (4).⁴

²We use the term ‘morpheme’ rather than ‘affix’ here to remain faithful to the terminology of Embick and Marantz (2008). In our implementation, we take morphemes to be syntactic terminals consisting of syntactic and semantic information and Vocabulary Items as exponents of such terminals following the standard Distributed Morphology assumptions (Embick & Marantz, 2008; Halle & Marantz, 1993). We take affixes to be some bound forms that can consist of one or more Vocabulary Items. Affixes have no real status in the theory (we assume) and they are just pre-theoretical descriptions of bound forms.

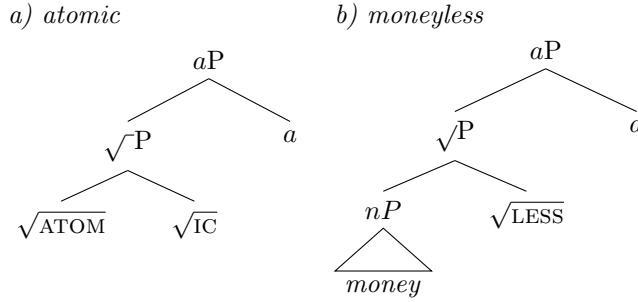
³As Marantz (2013) notes, the literature conflates the notion of polysemy resolution associated with heads and roots with special idiosyncratic interpretations. We take special idiomatic interpretations to be different from polysemy resolution, and following Marantz (2013), we also assume that they are domain sensitive. This point will be relevant in Section 4.4. We thank the anonymous reviewer who brought this to our attention.

⁴Some other affixes listed by Lowenstamm (2014) are *-able*, *-al*, *-an*, *-ant*, *-ary*, *-ate*, *-ible*, *-ic*, *-ive*, *-ory*, *-ous*, *-esque*, *-y*, *-ish*, *-en*, *-ful*. See Lowenstamm (2014, 8) for examples and further details.

- (4) English *-ian* suffix is categorially flexible (Lowenstamm, 2014, 2)
- a. *librar-ian* → NOUN
 - b. *reptil-ian* → ADJECTIVE

De Belder (2011) and Lowenstamm (2014) proposed that derivational affixes are best captured in a theory where they are bound roots that can combine with other roots or phrases as in (5). Category information is then added later by functional category-defining heads.

- (5) derivational morphemes as roots

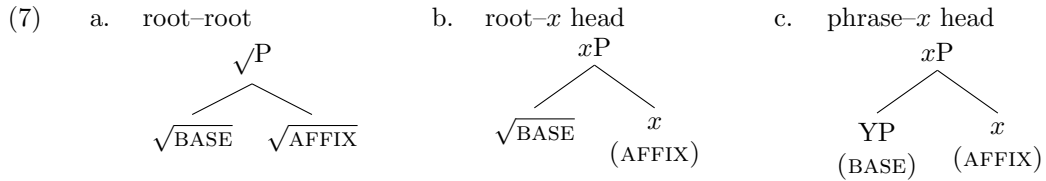


In this model, the distinction between stress-shifting and non-stress-shifting affixes is captured by whether the affix combines with another root or a phrase. This corresponds to the Level I versus Level II affixes of Kiparsky (1982).

Recently, Creemers et al. (2018) proposed an alternative middle ground where some derivational affixes are roots and others are category-defining functional heads. Their main argument follows from the observation that only a small subset of derivational affixes are categorially flexible. For example, in Dutch, De Belder (2011) reports that only 20% of the affixes are flexible. This is unexpected in a theory where all the derivational affixes are treated as roots. Instead, they propose (6).

- (6) Only affixes showing categorial flexibility are roots; all other derivational affixes realize functional $[x]$ heads. (Creemers et al., 2018, 48)

The proposal in (6) predicts three possible combinations of an affix with its base as shown in (7).



Creemers et al. (2018) show that the three types of combinations in (7) correspond to the three types of derivational affixes observed in Dutch with the following properties. Affixes with the structure in (7-a) are categorially flexible as they do not

introduce any category information, and the category of the $\sqrt{P}$ is defined by a higher functional head. In addition, direct combination of the affix with the base puts them into the same spell-out cycle leading to Level I phonological processes such as stress-shifting. Affixes in (7-b) are categorially rigid as they are functional heads and they do introduce category information. However, they still show Level I properties as they are spelled out along with the base under the assumption that phases only spell out phasal/cyclic complements (Embick, 2010). Finally, affixes with the structure in (7-c) are neither categorially flexible nor stress-shifting, the latter of which follows from the fact that they combine with a phasal/cyclic complement (a phrase that has already been categorized) and are thus in distinct spell-out domains than their bases. In short, Creemers et al. (2018) proposed a theory where some derivational affixes are roots while others are functional heads. This proposal leads to three types of affixes and two key tests used in classifying affixes are *categorial flexibility* and *stress-shifting*.⁵

So far, we presented various views on the root versus functional head status of derivational affixes. Another dimension to consider regarding the nature of affixes is their mode of attachment. Some affixes combine with their bases as heads, determining the category and grammatical properties of the word via projection. Others combine with their bases as modifiers, contributing lexical information but no category information or other grammatical features.

Wiltschko and Steriopo (2007) observed that the diminutive affixes in German and Halkomelem differ in systematic ways. In German, the diminutive suffix *-chen* always produces nouns of neutral gender regardless of the gender of the base (8). In addition, diminutivized nouns always behave like count nouns even though their bases can be mass nouns (9).

- (8) German: Change in gender (Wiltschko & Steriopo, 2007, 2)
- a. der Baum – das Bäum-chen
DET.MASC tree – DET.NEUT tree-DIM
'tree' – '(cute) little tree'
 - b. die Flasche – das Fläsche-chen
DET.FEM bottle – DET.NEUT bottle-DIM
'tree' – '(cute) little bottle'
- (9) German: Change from mass to count (Wiltschko & Steriopo, 2007, 2)
- a. viel Brot – viele Bröt-chen
much bread – many bread-DIM
'much bread' – 'many (cute) little sandwiches'

Unlike in German, diminutivization in Halkomelem does not change the grammatical properties of the base. In Halkomelem, nouns, verbs, and adjectives can be diminutivized. However, diminutivization does not change the category of bases (10). Nor does it turn mass nouns into count nouns (11).

⁵Creemers et al. (2018) use two other tests to help identify the nature of an affix. These are i) the ability of an affix to combine with a bound root and ii) relative order of affixes with respect to the base and other affixes. See Creemers et al. (2018) for details.

- (10) Halkomelem: No change in category (Wiltschko & Steriopo, 2007, 3)
- a. q'á:mi – q'á-q'emi
girl – DIM-girl
'girl' – 'small girl' N - N
 - b. lhí:m – lhi-lhi:m
picking – DIM-picking
'picking' – 'picking a little bit' V - V
 - c. p'eq' – p'í-p'eq'
white – DIM-white
'white' – 'a little white, whitish' A - A
- (11) Halkomelem: No change in mass property (Wiltschko & Steriopo, 2007, 3)
- a. s-peháls – s-pi-peháls
NOM-wind – NOM-DIM-wind
'wind' – 'a little bit of wind/breeze' No individuated reading

Wiltschko and Steriopo (2007) account for the difference in the behavior of German and Halkomelem diminutive suffixes outlined above as a matter of their mode of attachment. In German, the diminutive suffix is attached to its base as a head, and thus projecting its own features. This results in a change from the base to the derived word in terms of grammatical features such as gender and count/mass information. In contrast, in Halkomelem, the diminutive attaches to its base as an adjunct. Since the adjunct is not projected, no changes in the grammatical properties such as syntactic category or count/mass distinction are observed.

In a cross-linguistic study, Fábregas (2013) analyzed diminutives and proposed that some diminutives are heads, while others merge to their bases as specifiers. The distinction accounts for a range of syntactic, semantic, and phonological properties such as changing syntactic category, inducing special semantic interpretations, and productivity, among others. See Fábregas (2013) for details and Manova, Grestenberger, and Korecky-Kröll (2023) for a summary.

In a similar line of research, Gouskova and Bobaljik (2022) analyze the baby diminutive suffix (-*onok*) in Russian, and they argue that it can attach to its base both as a head and a modifier. When the diminutive suffix attaches as a head, it determines the gender of the word, triggers root suppletion, and gets to be interpreted as a baby diminutive (e.g., cat – kitten). On the other hand, when it attaches as a modifier, it does not affect the gender of the word or cause any suppletion. Plus, it is interpreted as an evaluative rather than a baby diminutive. Note that what Fábregas and Marín (2018) terms *specifier* corresponds essentially to what Gouskova and Bobaljik (2022) call a *modifier*: in both cases, the diminutive contributes evaluative meaning but does not project morphosyntactic features such as gender or declension class.⁶

To sum up this section, we have laid out various views and questions on the syntactic nature of derivational affixes: i) Are they roots or functional heads? and ii) Are they heads or modifiers? Table 2 provides a list of tests mentioned in this section in determining these properties of affixes.

⁶See also Wiltschko and Steriopo (2007) for a relevant account of various Russian diminutive suffixes.

Table 2 Summary of Diagnostics

Test	Root vs. Functional Head	Head vs. Modifier
Categorial Flexibility	Categorially flexible affixes are roots.	—
Stress-shifting	Stress-shifting affixes can be roots or category-defining functional heads. If latter, it has to be the first category-defining functional head.	—
Gender Changing	Functional Head (Probably)	Head
Category Changing	Functional Head	Head
Mass/Count Changing	Functional Head (Probably)	Head

The main essence of all the tests discussed above can be reduced to three core properties: 1) bearing category information, 2) projection, and 3) locality. Morphemes that bear category information are functional heads, others are roots. Morphemes that project (e.g., category, gender, mass/count information, etc.) are heads, the ones that do not project are modifiers (adjuncts). Finally, morphemes that are ‘close-enough’ to the base (i.e. in the same spell-out domain) are potentially stress-shifting.⁷ In the next section, we discuss the properties of diminutives in Turkish in the light of some of the applicable tests mentioned above and some others before we move on to our main proposal.

3 The Diminutive Suffix in Turkish

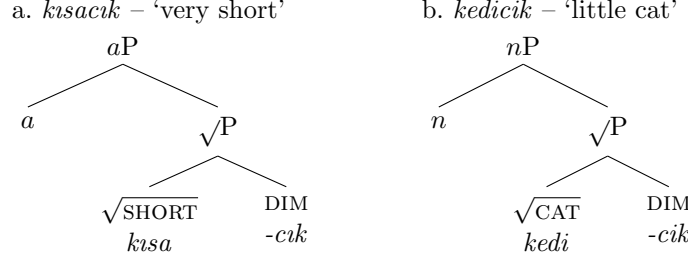
This section lays out the properties of the diminutive suffix *-Cik* in Turkish. *-Cik* has been analyzed as a diminutive marker (Göksel & Kerslake, 2004; Sebüktekin, 1984; Sezer, 1981; Taylan, 2015; Zimmer, 1970) and as an augmentative marker (Türk, 2020). It combines with a range of syntactic categories including adjectives, nouns, demonstratives, and quantificational expressions. These are illustrated in (12).

- (12)
- a. ince – ince-cik
thin – thin-DIM
‘thin – very thin’ adjective
 - b. ayı – ayı-cık
bear – bear-DIM
‘bear – little/cute bear’ common noun
 - c. Zeynep – Zeynep-çik
Zeynep – Zeynep-DIM
‘Zeynep – little/dear Zeynep’ proper noun
 - d. bura – bura-cık
here – here-DIM
‘here – right here’ demonstrative
 - e. bir – biri-cik
one – one-DIM
‘one – dear one’ numeral

⁷Some other relevant effects of locality are *root suppletion*, selectional restrictions, ‘special’ meanings, etc. These will be discussed in the following sections where relevant.

As observed in (12) *-Cik* appears to be flexible in terms of what it combines with and also flexible in terms of what it returns. Thus, given the criteria defined in Creemers et al. (2018), *-Cik* is predicted to be a root rather than a functional head. The predicted representations are given in (13).

(13) Predictions of Creemers et al. (2018) for flexible affixes



We argue that the apparent categorial flexibility of *-Cik* is illusory, and that categorial flexibility alone is not a reliable diagnostic.⁸ Underlyingly, *-Cik* is a category-defining functional head that combines with a root and always returns an adjective. The structure of adjectival diminutives formed with *-Cik* is in line with the structures in (3), yet, the structure of nominal diminutives is significantly different. We postpone the discussion until Section 4 as it requires elucidation of our key assumptions. Throughout the paper, we mainly focus on adjectives and common nouns. We believe that our analysis can easily extend to other categories. In the following, we start with the properties of adjectival diminutives, which form the basis of our core analysis, and then move on to the discussion of nominal diminutives.

3.1 Adjectival Diminutives

Adjectival diminutives are formed by combining an adjectival root with the suffix *-Cik*. One of the key properties of *-Cik* is that it can only combine with gradable adjectives as in (14) while it is not compatible with non-gradable adjectives as shown in (15).

(14) *Gradable Adjectives*

- a. kısa - kısa-cık
short - short-DIM
‘short’ - ‘very short’
- b. taze - taze-cık
fresh - fresh-DIM
‘fresh’ - ‘very fresh’

(15) *Non-gradable Adjectives*

- a. dolu - *dolu-cuk
full - full-DIM
‘full’ - ‘very full (intended)’

⁸See Oğuz (2024) who considers *-Cik* to be a categorially flexible affix, based on a previous version of our manuscript. We revised our previous analysis in this version based on feedback from an anonymous reviewer.

- b. açık - *açık-cık
open - open-DIM
'open' - 'very open (intended)'

Among gradable adjectives Bierwisch (1989) makes a distinction between dimensional and evaluative adjectives. Dimensional adjectives measure some physical property of an object along some dimension (e.g., height, size, volume, etc.) whereas evaluative adjectives measure some subjective property (e.g., beauty, cuteness, expensiveness, etc.). *-CIk* only combines with dimensional adjectives as shown in Table 3.

Table 3 *-CIk* only combines with dimensional adjectives.

Dimensional Adjectives			Evaluative Adjectives		
English	Turkish	Diminutive	English	Turkish	Diminutive
short	kısa	kısacık	beautiful	güzel	*güzecik
thin	ince	incecik	ugly	çirkin	*çirkincik
small	küçük	küçücük	cheap	ucuz	*ucuzcuk
tiny	minik	minicik	expensive	pahalı	*pahalcık

A key property of *-CIk* is that it can only combine with dimensional adjectives that make reference to the lower end of a scale. Assuming that adjectives like *kısa* – *uzun* ('short' – 'tall') correspond to the lower and upper ends of a HEIGHT scale, as depicted in Figure 1, *-CIk* only combines with adjectives that are on the lower end. Table 4 illustrates this with examples from both ends of the HEIGHT scale.

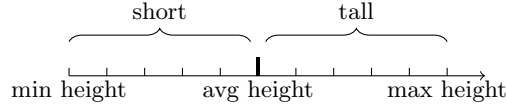


Fig. 1 Lower and upper ends of the HEIGHT scale.

Table 4 *-CIk* only combines with adjectives on the lower end of a scale.

Dimension	Adjective	Lower End			Upper End	
		Regular	Diminutive		Regular	Diminutive
VOLUME	small	küçük	küçücük	big	büyük	*büyücük
VOLUME	little	minik	minicik	big/huge	kocaman	*kocamancık
LENGTH	short	kısa	kısacık	tall	uzun	*uzuncuk
HEIGHT	low	alçak	alçacık	high	yüksek	*yüksecik
WIDTH	narrow	dar	daracık	wide	geniş	*genişçik
WIDTH	thin	ince	incecik	thick	kalın	*kalıncık
AGE	young	genç	gencecik	old	yaşlı	*yaşlıcık
AGE/TIME	fresh	taze	tazecik	stale	kart/bayat	*kartcık/bayatcık
HARDNESS	soft	yumuşak	yumuşacık	hard	sert	*sertçik

Although *-CIk* only combines with dimensional gradable adjectives that make reference to the lower end of a scale, it is not completely productive and cannot combine

with all such adjectives. For example, it can combine with some time or age related adjectives but not others, as illustrated in Table 5.

Table 5 Choice of what scale $-Cik$ combines with is idiosyncratic.

English	Turkish	Diminutive	English	Turkish	Diminutive
young	genç	gencecik	cold	soğuk	*soğu(k)cuk
fresh	taze	tazecik	cool	serin	*serincik
little	ufak	ufacık	slow	yavaş	*yavaşcık
			new	yeni	*yenicik
			near	yakın	*yakıncık ¹

¹*yakıncık* is OK when it is used as a place name (e.g., a village). See Appendix I for more details on diminutivized place names.

To sum up the selectional properties of adjectival diminutives, we observe that *-Cik* only combines with a subset of dimensional gradable adjectives that are only on the lower end of a scale. Especially, the fact that it never combines with adjectives that are on the higher end of a scale will play a key role in our analysis in Section 4.⁹

A crucial point to note is that adjectival diminutives convey an intensified reading on the adjectives that are on the lower end of a scale. For example, *küçücük* ‘very small’ has an intensified smallness meaning.¹⁰ One way to view this intensification is the minimization of a scale. Thus, *küçücük* denotes some contextually determined minimal degree on a size scale. In Section 4, we build a compositional semantic analysis of the diminutive morpheme based on this intuition. Now that we have described the selectional properties of *-Cik*, we continue with its morphophonological properties.

The first morphophonological property to note is that *-CIk* is a stress-shifting affix. [Özçelik \(2023\)](#) describes it as a pre-pre-stressing suffix. Regular stress in Turkish is word final and outer suffixes regularly attract stress ([Göksel & Kerslake, 2004](#); [Lewis, 1967](#); [Özçelik, 2023](#); [Sezer, 1981](#); [Underhill, 1976](#)). This is shown in (16). However, when *-CIk* combines with an adjective, it shifts stress to two syllables before the suffix itself. This is shown in Table 6.

- (16) Regular word-final prominence (Özçelik, 2023, 498)
- a. dö^ˈv ‘beat’
- b. dö^ˈv.dür ‘have (someone) beaten’
- c. dö^ˈv.dürt.tü ‘had (someone) have (someone) beaten’
- d. dö^ˈv.dürt.tü.lər ‘they had someone have him beaten.’

⁹One exception to this generalization is *sıcak – sıcaık* ‘warm – pleasantly warm’. See Section 5 for a brief discussion on *sıcaık*.

¹⁰This is why Türk (2020) treated it as an augmentative marker rather than a diminutive marker when it combines with an adjectival base. See Section 4.1.1 for an explanation of why we do not pursue this line of analysis.

Table 6 Stress patterns in adjectival diminutives

English	Turkish	Diminutive
short	kı.sa ^ˈ	kı.sa.cık ^ˈ
thin	in.ce ^ˈ	in.ce.cık ^ˈ
small	kü.çük ^ˈ	kü.çü.cük ^ˈ
tiny	mi.nik ^ˈ	mi.ni.cık ^ˈ

The second morphophonological property of *-Cik* emerges when it combines with a monosyllabic adjective. In this context, an epenthetic vowel appears between the adjectival base and the *-Cik* suffix as illustrated in Table 7.

Table 7 Vowel Epenthesis in Monosyllabic Roots

English	Turkish	Diminutive
narrow	dar	dar- a -cık
young	genç	genc- e -cık
one	bir	bir- i -cık
little	az	az- ı -cık

The third and final point is that *-Cik* triggers the deletion of the root-final [k] in the adjective, as illustrated in 8.

Table 8 [k] deletion in adjectival diminutives

English	Turkish	Diminutive
small	küçük	küçü-cük
tiny	minik	mini-cık
little	ufak	ufa-cık
low	alçak	alça-cık
hot	sıcak	sıca-cık
soft	yumuşak	yumuşa-cık

In a nutshell, when the diminutive affix *-Cik* combines with an adjective, it denotes a minuscule degree on the scale associated with the adjective. This is correlated with the fact that it can only attach to dimensional gradable adjectives that are on the lower end of a scale (e.g., *küçük* ‘small’ but not *büyük* ‘big’). Even among these adjectives, there is a degree of idiosyncrasy in the sense that *-Cik* does not combine with all such adjectives. Besides these properties, *-Cik* displays some Level I phonological properties by shifting the stress to two syllables before itself, causing the epenthesis of a vowel in monosyllabic adjectives, and finally deleting the final [k] on the adjectival root. All of these properties point out to root proximity. We take it that *-Cik* attaches directly to the adjectival root in all these instances. In the next section, we describe the properties of nominal diminutives formed with *-Cik*.

3.2 Nominal Diminutives

Nominal diminutives in Turkish are formed by combining a nominal base with the *-Cik* suffix. The process is productive and yields the meaning of ‘small’ and (usually) ‘endearing’. The absolute size of the entity denoted by the noun is irrelevant. Instead, it is the relative size that matters. For example, cats, elephants, giraffes can all be diminutivized as in (17).

- (17) a. kedi-cik
cat-DIM
‘little/lovely cat’
b. fil-cik
elephant-DIM
‘little/lovely elephant’
c. zürafa-cık
giraffe-DIM
‘little/lovely giraffe’

The endearment meaning is not entailed by the diminutive suffix itself but may become conventionalized when certain nominals combine with it. For instance, *kedicik* is frequently used as a term of endearment. With other nouns, however, the diminutive may give rise to different conventionalized readings, such as belittlement; and in still other cases, no special conventionalization occurs, leaving only the literal ‘very small’ meaning. Consider (18) and (19).

- (18) insan-cık-lar
human-DIM-PL
‘little humans’ (used in a belittling context)
(19) Bu ev değil, ev-cik. Hiç beğen-me-di-m.
this house NEG house-DIM any like-NEG-PST-1.SG
‘This is not a (regular-sized) house, but a very small one. I hated it.’

-Cik can be productively combined with any noun provided that the referent can plausibly undergo diminutivization. The semantic contribution of *-Cik* is transparent except in certain words where the diminutivized noun takes a special idiomatized interpretation in addition to its transparent diminutive meaning.¹¹ These are illustrated in Table 9.

Morphophonologically, *-Cik* on nominal bases displays a range of differences from *-Cik* on adjectival bases. The first difference regards stress assignment. While on adjectival bases, *-Cik* is pre-pre-stressing, on nominal bases, stress assignment is regular (i.e. word final). This is illustrated in 10.

In addition to the difference in stress patterns, nominal diminutives do not lead to vowel epenthesis with monosyllabic bases. Nor do they display [k] deletion.¹² Table 11 illustrates these properties.

¹¹See Manova and Sitchinava (2023) for a discussion on the ubiquity of special idiomatized readings in diminutives and similar examples from Slavic.

¹²There are some dialects where [k] deletion is observed with some nominal diminutives such as *köpek-köpecik* ‘dog-little doggie’, *ayak-ayacık* ‘foot-little foot’. We believe these dialects must have developed a

Table 9 Special idiomatic interpretations in diminutive nouns

English	Turkish	Diminutive	Transparent Meaning	Idiomatic Meaning
lid	kapak	kapak-çık	little/lovely lid	valve (e.g., heart valve)
ear	kulak	kulak-çık	little/lovely ear	paddle (e.g., a car gear paddle)
barley	arpa	arpa-çık	little/lovely barley grain	stye
book	kitap	kitap-çık	little/lovely book	leaflet
söz	expression	söz-cük	little expression	word

Table 10 Regular stress in nominal diminutives: Stress is on the last syllable.

English	Turkish	Diminutive	Plural Diminutive
cat	ke.di ^ˈ	ke.di.ci ^ˈ k	ke.di.ci ^ˈ k.lər
car	a.ra.ba ^ˈ	a.ra.ba.ci ^ˈ k	a.ra.ba.ci ^ˈ k.lər
ant	ka.rın.ca ^ˈ	ka.rın.ca.ci ^ˈ k	ka.rın.ca.ci ^ˈ k.lər
box	ku.tu ^ˈ	ku.tu.cu ^ˈ k	ku.tu.cu ^ˈ k.lər

Table 11 Epenthesis and [k] deletion not available in nominal diminutives

English	Turkish	Diminutive	Unavailable
elephant	fil	fil-cık	* fil-i-cık
mountain	dağ	dağ-cık	* dağ-ı-cık
lake	göl	göl-cük	* göl-ü-cük
horse	at	at-çık	* at-ı-cık
dog	köpek	köpek-cık	*köpe-cık
ear	kulak	kulak-cık	*kula-cık
lid	kapak	kapak-cık	*kapa-cık
leaf	yaprak	yaprak-cık	*yapra-cık

So far, we have presented the facts regarding the distribution and properties of *-Cık* on adjectival and nominal bases. Table 12 summarizes all the properties in a nutshell.

4 Proposal

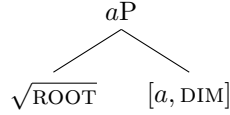
In this section, we lay out our core assumptions, the main proposal, and its detailed implementation within the Distributed Morphology framework. We propose that *-Cık* is the realization of the category-defining functional head DIM. DIM introduces the diminutive semantics as well as the adjectival category. Adjectival diminutives are formed by combining DIM directly with the adjectival root, turning it into a diminutivized adjective. This is illustrated in (20). Nominal diminutives, in contrast, are formed by combining an *nP* with an adjectival phrase whose adjectival root is an abstract magnitude scale with a null exponent. This is illustrated in (21).

regular phonological [k]-deletion rule extrapolating the adjectival diminutive patterns. Nevertheless, we do not have sufficient data to precisely describe the behavior, and hence leave it for further research.

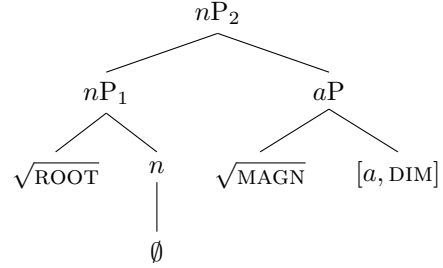
Table 12 Properties of *-Clk* on adjectival and nominal bases

	adjectival diminutive	nominal diminutive
selection	dimensional gradable adjectives on the lower end of a scale	any nominal base
meaning	intensify the degree denoted by the adjective	size diminutive (little), endearment & belittling readings possible
stress-shifting	pre-pre-stressing	regular word final
epenthesis	Yes	No
[k] deletion	Yes	No
idiomatic reading	Not attested	Yes (on some nouns)

(20) adjectival diminutive



(21) nominal diminutive



In the following, we start by outlining our core theoretical assumptions. Then, we build an account of adjectival diminutives on these assumptions in light of the data we presented in Section 3. Finally, we provide an account of nominal diminutives.

4.1 Assumptions

Adopting Distributed Morphology (Embick & Marantz, 2008; Halle & Marantz, 1993) as our framework, we assume that morphemes are syntactic terminals consisting of syntactic and semantic information but no phonological content. Morphemes receive their phonological content through an exponence process called Vocabulary Insertion (Halle & Marantz, 1993). We further assume that the well-formedness of a linguistic expression depends on the satisfaction of the syntactic, semantic, and morphophonological requirements on the morphemes and their exponents. Thus, a linguistic expression whose derivational history satisfies all the semantic and syntactic requirements but fails the morphophonological requirements is ill-formed. Likewise, a linguistic expression whose derivational history satisfies all the syntactic and morphophonological requirements but fails the semantic requirements is also ill-formed. Our unified account of adjectival and nominal diminutives rests on this very assumption.

4.1.1 *-Clk* is a category-defining functional head.

In Section 3, we showed that [k] deletion occurs in adjectival diminutives but not in nominal diminutives. To explain this contrast, we begin by examining the structure

of adjectives and their diminutive forms more closely. In what follows, we outline our core assumptions about the structure of adjectives that take the diminutive suffix and clarify the status of the diminutive morpheme. We begin with a detailed look at the structure of adjectives.

Türk (2020) observed that some of the gradable adjectives in Turkish end with [k] and this [k] disappears when these adjectives are turned into causative verbs as shown in Table 13. He took this as evidence to argue that [k] is not part of the adjectival base but an added morpheme on top of the adjectival base.

Table 13 [k] is not part of the adjectival base/root (Türk, 2020, 158)

adjective	English	causative verb	English
alça-k	low	alça-lt	lower
yumuşa-k	soft	yumuşa-t	soften

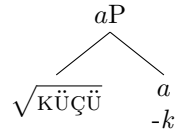
Table 14 extends Türk’s observations to inchoatives which serve as the base for causatives. The data show that adjectives alternate between *-k* and $\emptyset$ exponents while the inchoative verbs with the same bases alternate between *-l* and $\emptyset$ exponents.

Table 14 *-k* is an allomorph of the *a* head.

gloss	adjective	inchoative	causative
small	küçü-k	küçü-l	küçü-l-t
big	büyük	büyük- $\emptyset$	büyük- $\emptyset$ -t
low	alça-k	alça-l	alça-l-t
high	yükse-k	yükse-l	yükse-l-t
soft	yumuşa-k	yumuşa- $\emptyset$	yumuşa- $\emptyset$ -t
little	az- $\emptyset$	az-al	az-al-t
narrow	dar- $\emptyset$	dar-al	dar-al-t
thin	ince- $\emptyset$	ince-l	ince-l-t

Given the data in Table 14, we follow Türk (2020) in treating adjectives above to be complex. We argue that these adjectives minimally consist of a root $\sqrt{\text{ }}$ and a category-defining functional head *a*.¹³ We further argue that *-k* is the realization of the category-defining head *a*. As an illustrative example, (22) presents the structure of the adjective *küçük* ‘small’.

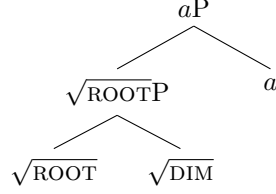
(22) Structure of *küçük* ‘small’



¹³It should be noted that *küçük*, *alça*, and *yükse* are bound roots, further supporting the assumption that adjectives ending in *-k* are complex.

Second, analyzing DIM as a category-defining head also accounts for the absence of diminutive verbs in Turkish formed with the diminutive suffix *-Cik* followed by the inchoative suffix *-l*. To appreciate this point, consider a plausible alternative analysis in which DIM is a root that combines with another root, with a null category-defining head. This alternative structure is illustrated in (26).

(26) A plausible hypothesis: DIM as a root. (Not adopted)



This alternative account predicts the possibility of finding inchoative verbs listed in Table 15.¹⁵ Yet, these are ungrammatical.¹⁶

Table 15 Predictions of DIM as a root.

gloss	adjective	diminutive adjective	diminutive verb
small	küçük-k	küçük-cük	*küçük-cük-ül
low	alça-k	alça-cık	*alça-cık-ıl
little	az-Ø	az-ı-cık	*az-ı-cık-ıl
narrow	dar-Ø	dar-a-cık	*dar-a-cık-ıl
thin	ince-Ø	ince-cık	*ince-cık-ıl

So far, we have argued that the diminutive suffix *-Cik* is the realization of the head DIM, which is a category-defining functional head that combines with a root to turn it into an adjective while also adding the diminutive meaning. We have also argued that adjectives ending in [k] are complex consisting of a root and the category-defining head *a* and *-k* realizes the *a* head. This accounts for the [k]-deletion in adjectival diminutives described in Section 3 while also accounting for the fact that [k]-deletion does not occur in nominal diminutives.

In Section 3, we presented a range of morphophonological properties unique to adjectival diminutives and distinct from nominal diminutives such as stress shift and vowel epenthesis. Eventually, we show that the analysis provided above accounts for all these properties. However, we postpone their discussion and take a detour to delve into the semantic properties of the diminutive morpheme DIM which paves the way

¹⁵We thank Asia Pietraszko for pointing this out.

¹⁶It is possible to express a change of state with a diminutivized adjective. To form such verbs, Turkish uses the suffix *-LAş* which combines with an adjective or a noun to express a change of state. This suffix though attaches to already categorized roots. Some illustrative examples are given below.

- (i)
- a. *küçük – küçükleş* ‘small – to become small’
 - b. *küçücük – küçücükleş* ‘very small – to become very small’
 - c. *köpek – köpekleş* ‘dog – to become dog’
 - d. *ışıl-l-tı-l – ışı-l-tı-l-laş* ‘sparkling – to become sparkling’

Example (i-d) is particularly illustrative as it shows that *-LAş* is attached to already categorized roots.

for articulating the syntactic structure of nominal diminutives in Turkish. The rest follows from the structural properties of the two types of diminutive constructions.

Before closing this section, a couple notes are due on two alternative hypotheses regarding the analysis of the suffixes *-k* and *-Cik*. First, [Türk \(2020\)](#) treated *-Cik* as a complex suffix consisting of *-CI* and *-k*. In earlier versions of this manuscript, we also attempted a split where *-CI* was treated as the diminutivizing root and *-k* as the adjectivizer head. However, this analysis makes the wrong prediction that we should expect to see diminutive verbs, which we abandon in the current version.

The second and more serious issue regards the root versus functional head status of the affix *-k*. As an anonymous reviewer notices, Turkish has a number of nouns that are derived from verbs and also end with *-k*. Some such nouns are listed in [Table 16](#).

Table 16 *-k* deriving nouns

verb gloss	verb	noun	noun gloss
wish	dile	dile- k	wish
lay	döşe	döşe- k	mattress
sift	ele	ele- k	sieve
close	kapa	kapa- k	lid
emit light	ışıl	ışıl- k	light
recognize	tanı	tanı- k	witness

In all the examples in [16](#), the forms that bear *-k* are nouns and crucially they do not have any adjectival status. They are all derived from bases where the base forms can be used as verbs without any additional morphology.¹⁷

The anonymous reviewer also points out another set of words that bear [k]. These are also derived from bases that can be used as verbs without any additional morphology. What is different in this set is that the words can be used as nouns or adjectives.

Table 17 *-Ik* deriving nouns and adjectives

verb gloss	verb	word	gloss
sink	bat	bat- ık	sunk (<i>adj.</i>), shipwreck (<i>n.</i>)
turn	dön	dön- ük	turned toward (<i>adj.</i>)
flee	kaç	kaç- ık	eccentric (<i>adj.</i> , <i>n.</i>)
alight	kon	kon- uk	guest (<i>n.</i>)
scatter	saç	saç- ık	scattered (<i>adj.</i>)

Building on the examples presented and drawing from [Erdal \(1991\)](#) and [Karaoğlu \(2022\)](#), the anonymous reviewer observes that the Old Turkic etymon of the suffix *-k* is *-(O)k*. They argue that in all the relevant cases, the modern reflex is *-(I)k*,

¹⁷If *-k* in these examples is added to roots, then it must be an accident that all of these roots combine with a null verbalizer to be used as a verb.

which displays categorical flexibility—appearing in both nominal and adjectival contexts. Consequently, the reviewer suggests that this suffix should be analyzed not as a category-defining functional head, but rather a root.

Although we appreciate the reviewer’s insights, we do not pursue this line of analysis for two reasons. First, the primary aim of this paper is to account for the properties of the diminutive suffix *-Cİk*, not *-k*. A detailed investigation of the status of *-k* falls beyond the scope of the present study. In addition, as the reviewer notes, our main proposal concerning the distinction between adjectival and nominal diminutives is compatible with either analysis of *-k*. In our proposal, the absence of *-k* in adjectival diminutives follows from competition for the same structural position as the DIM head. Under an alternative analysis, where *-k* is the realization of a root attaching below the category-defining functional head, its absence may be attributed to outward-sensitive allomorphy triggered by the presence of a local DIM head. Such locality conditions do not hold in nominal diminutives, where *-k* surfaces as expected. See Section 4.3 for details.

Second, we are not fully convinced that all the words ending in [k] as listed above are derived by the same morpheme and have the same underlying structure. While the derived words in Table 17 are either flexible between noun and adjective categories or they are at least coercible, the ones in Table 16 are not. If these were derived by the same root, we would expect to see the words *dilek*, *elek*, *ışık*, *döşek*, *tanık*, etc. to be available as or at least coercible into adjectives. None of them are. Turning them into adjectives requires other affixes as listed below in Table 18.

Table 18 nouns derived with *-k* and their corresponding adjectives

verb gloss	verb	noun	noun gloss	adj	adj. gloss
lay	döşe	döşe- k	mattress	döşe-li	paved/layed/tiled
close	kapa	kapa- k	lid	kapa-lı	closed
emit light	ışI	ışI- k	light	ışık-lı	lit
recognize	tanı	tanı- k	witness	tanı-dık	recognized

Another minor but important detail concerns the form of these suffixes. There is reason to believe that *-k* and *-(I)k* are not simply variants of the same suffix. If the reviewer is correct in treating all these suffixes as morphophonological variants of *-(I)k*, then roots ending in a vowel should not appear with *-(I)k*; rather, they should appear with *-k*. However, there is one verb ending in a vowel that takes *-(I)k* plus an epenthetic *y* between the root and the suffix. This verb is *eri* ‘melt’, whose adjectival form is *eri-yik* ‘molten’, not **eri-k*. Unfortunately, this is the only example we have found, and few verbs ending in a vowel combine with either *-k* or *-(I)k* to form adjectives. Nevertheless, this example suggests a potential distinction between *-k* and *-(I)k*, to the extent that the available data allow.

To close this section, we have argued that *-k* realizes the category-defining head *a* and *-Cİk* realizes the category-defining head DIM. *-k* is lost in adjectival diminutives as *a* and DIM are two flavors of the same category-defining head that creates adjectives by combining with a root. Our analysis is also compatible with an alternative view in

which $-k$ is a root rather than a category-defining head. However, we do not pursue this alternative here and leave it for future research, as we currently lack sufficient data to adjudicate between these possibilities, and it does not constitute the central issue addressed in this paper.

4.2 The diminutive morpheme is a minimizer.

In Section 3, we have shown that the diminutive morpheme DIM is only compatible with the set of dimensional gradable adjectives that are on the lower end of a scale (with some idiosyncratic exceptions). What is striking is that it does not combine with an adjective that makes reference to the higher end of a scale.¹⁸ For example, *küçük* ‘diminutive small’ is possible whereas *büyük* ‘diminutive big’ is not. Table 4 provides more examples.

Informally, we define DIM as a minimizer. It picks out a degree *maximally smaller than* a contextually defined standard associated with an adjective. This is illustrated for *kısa* ‘short’ in Figure 2.

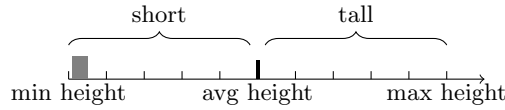


Fig. 2 Degree denoted by *kısa* ‘diminutive short’

Our informal definition captures the intended meaning of a diminutivized adjective but it makes the wrong prediction. If the diminutive morpheme is a minimizer that attaches to adjectives, then it should be able to combine with gradable adjectives on the higher end of the scale as well. For example, we should be able to form **uzuncuk* ‘diminutive tall’ which is graphically illustrated in Figure 3. Compositionally, this is a well-formed interpretation as it simply means the smallest degree above the average (e.g., ‘the shortest among the tall’).

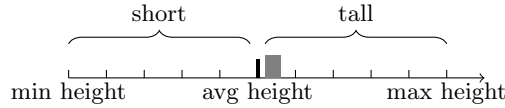


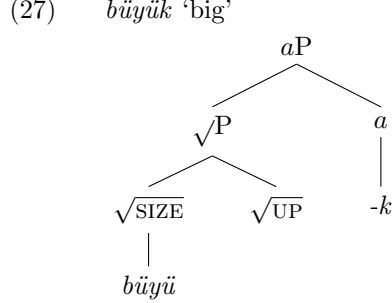
Fig. 3 Hypothetical degree denoted by **uzuncuk* ‘diminutive tall’

We argue that the diminutive morpheme directly combines with the entire scale and always returns the minimum non-zero degrees associated with the scale. This is why it seems to combine with adjectives on the lower end of a scale but not the ones on the higher end. In the following, we provide a detailed implementation of this idea.

Following Kennedy (2007, 2013); Kennedy and McNally (1999), we assume that gradable adjectives have measure functions in their denotations that measure the degree of an individual alongside a dimension such as height, weight or length etc.

¹⁸The only known exception is *sıcak* ‘hot/warm’. See Section 5 for a potential account.

Accordingly, we argue that the scalar roots in our representation denote functions from individuals to degrees (type $\langle e, d \rangle$). Their meanings are further modified by functions such as UP, DOWN, and DIM. An adjective like *büyük* ‘big’ will be composed of a height scale, the function UP and an adjectivizer head (*a*), as illustrated in (27). We assume that the function UP has the meaning defined in (28-b).



We define the meaning of the scalar root and the functor as in (28-a).

- (28) a. $\llbracket \sqrt{\text{size}} \rrbracket = \lambda x. \text{size}(x)$
b. $\llbracket \sqrt{\text{UP}} \rrbracket = \lambda g_{\langle e, d \rangle}. \lambda x. g(x) \succeq s(g)$, where s is a function that returns the standard degree associated with the measured property (Kennedy, 2007).

Based on the meanings we defined in (28-a), we can calculate the meaning of the $\sqrt{P}$ in (27).

(29) $\llbracket \sqrt{P} \rrbracket = \lambda x. \text{size}(x) \succeq s(\lambda x. \text{size}(x))$

An important note is in order here. Notice that the comparison class is not part of our syntactic representation. Instead, we follow Kennedy (2007) in using the function s in the meaning of UP to pick up a standard degree of height (possibly based on a contextually salient comparison class). Kennedy (2007) proposes that the effect of the comparison class in standard treatment could be achieved by restricting the domain of the scale function such that it only measures the degree for a specific group of individuals (e.g., $\lambda x : \text{car}(x). \text{size}(x)$).¹⁹

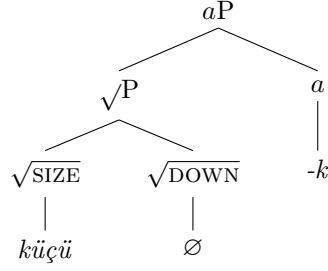
Similar to how UP combines with the scalar root, DOWN also combines with a scalar root and restricts its meaning further. However, unlike the former, DOWN expresses the condition that x ’s degree on a scalar dimension is below what is determined to be the standard degree for a comparison class in a given context, under the simple assumption that any degree below the standard corresponds to the lower end of a scale. We can therefore define its meaning as in (30).

(30) $\llbracket \text{DOWN} \rrbracket = \lambda g_{\langle e, d \rangle}. \lambda x. g(x) \prec s(g)$

¹⁹Alternatively, s can evaluate the standard relative to a contextually determined comparison class, without syntactic representation (e.g., $s(g, C) =$ the standard degree on the scale of g , as determined over the set C)

Similar to the structure proposed for *büyük* ‘big’, a gradable adjectival form like *küçük* ‘small’ is composed of a size root, a function and an adjectivizer as illustrated in (31).

(31) *küçük* ‘small’



Hence, the meaning of $\sqrt{P}$ is understood to be the characteristic function of the set of individuals whose sizes are smaller than what is determined to be the standard size as illustrated in (32).²⁰

$$(32) \quad \llbracket \sqrt{P} \rrbracket = \lambda x. \text{size}(x) \prec s(\lambda x. \text{size}(x))$$

We propose that the syntactic status of DIM is similar to UP and DOWN, except that its semantic contribution and syntactic categorization are conflated into a single head. Semantically, the function of DIM is closest to DOWN. Both require that an individual’s degree on a scalar dimension be *below the contextual standard* $s(g)$. However, DIM is not satisfied by any degree below $s(g)$; instead, it requires that the degree fall *within the lowest portion of the scale*, i.e. near its infimum. In other words, the diminutive expresses not just *smaller than standard* but *small enough to count as belonging to the bottom range of the scale*.

Formally, let $\text{Range}(g)$ be the set of contextually relevant degrees for dimension g . Let $\inf(\text{Range}(g))$ be its lower bound and $s(g)$ the contextual standard. We define a *cutoff point* θ that marks the top of the bottom range as follows:²¹

$$(33) \quad \theta = \inf(\text{Range}(g)) + \delta \cdot (s(g) - \inf(\text{Range}(g))), \text{ where } 0 < \delta \ll 1 \text{ is a contextually fixed proportion (for instance, the bottom 10–20\% of the scale).}$$

The *bottom range* $\text{BR}(g)$ is then:

$$(34) \quad \text{BR}(g) = [\inf(\text{Range}(g)), \theta].$$

Accordingly, the meaning of DIM can be defined as in (35):

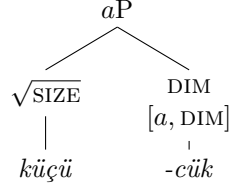
$$(35) \quad \llbracket \text{DIM} \rrbracket = \lambda g_{\langle e, d \rangle}. \lambda x. g(x) \in \text{BR}(g)$$

²⁰Fábregas and Marín (2018) propose a similar idea regarding Spanish adjectives being syntactically complex. While our implementation is orthogonal to theirs, it shares the same spirit. In a similar vein, Wyngaerd, Caha, and De Clercq (2024) analyze scalar adjectives to consist of a scalar root and some further heads in a Nanosyntactic framework.

²¹This definition abstracts away from how contextual values such as $\text{Range}(g)$ or $s(g)$ are determined in practice. The aim here is simply to capture the intuition that diminutives target a region in the lowest end of a scale.

An illustrative example of the proposed syntactic structure of a diminutivized adjective is given in (36).

(36) *küçücük* ‘very small’



Accordingly, the meaning of the *aP* in (36) is the characteristic function of a set of individuals whose sizes fall within the *bottom range* (BR) of the scale, i.e. the interval from the minimum of the scale up to a contextual cutoff θ below the standard degree of size for a comparison group.

(37) $\llbracket aP \rrbracket = \lambda x. \text{size}(x) \in \text{BR}(\text{size})$ ‘diminutive small’

On the semantic side, then, the diminutive always combines with the full scale and returns its lower range. What gives the impression that it combines only with lower end adjectives like *small* is in fact a matter of morphophonology: just as the functional head DOWN conditions the size root to surface as *küçü-*, the diminutive head triggers the same conditioning. As a result, we do not see forms like “diminutive-tall”, but only the conditioned *küçü-* root. In Section 4.4 we will show that DIM hosts the feature bundle $\{a, \prec, \text{br}\}$; the presence of $\prec$ is plausibly shared with DOWN, which makes the parallel conditioning effects unsurprising.

Our analysis makes a number of predictions. Since the diminutive form requires reference to the bottom range of the scale in its truth conditions, it will always entail its non-diminutive form. In other words, *küçücük* ‘very small’ is predicted to entail *küçük* ‘small’. The infelicity of a follow-up clause asserting the negation of *small* shows this much, as illustrated in (38).

(38) Bu fil küçü-cük, #ama küçük değil.
 this elephant small-DIM but small not
 ‘This elephant is very small, but not small.’

Second, notice that under our analysis, the diminutive function guarantees that x ’s degree is not just below the contextual standard but specifically located in the bottom segment of the scale. Crucially, this description is *interval-based*: the bottom range may include multiple degrees, so two individuals can both qualify as *küçücük* while still being comparable in size. Indeed, such follow-ups are not infelicitous in Turkish, as shown in (39).

(39) Şu iki bina da bu şehir-de-ki bina-lar-a kıyasla
 That two building also this city-LOC-PRO building-PL-DAT compared
 küçü-cük, fakat sağ-da-ki sol-da-ki-nden daha küçük/büyük.
 small-DIM but right-LOC-PRO left-LOC-PRO-ABL more small/big

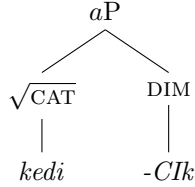
‘These two buildings both are very small compared to the buildings in this city, but the one on the right is smaller/bigger than the one on the left.’

4.3 Nominal diminutives are nouns modified by an adjectival diminutive

In the previous two sections, we have analyzed *-CIk* as the realization of a category-defining functional head that combines with a scale, which is a root, and returns an adjective with a diminutivized meaning. This was sketched in (36). Semantically, we defined the morpheme DIM as a minimizer that combines with a scale and returns the characteristic function of the set of individuals whose relevant scalar degrees are maximally smaller than what is considered to be the standard in the context of utterance.

Extending our analysis to nominals immediately runs into two problems, one syntactic and the other semantic. Let us start with an attempt to use the same analysis for nominals and show where it fails. Consider the nominal diminutive *kedicik* ‘little cat’ and an attempt to derive it by root-attaching the diminutive morpheme as in (40).

(40) *kedicik* ‘little cat’ (Wrong representation)



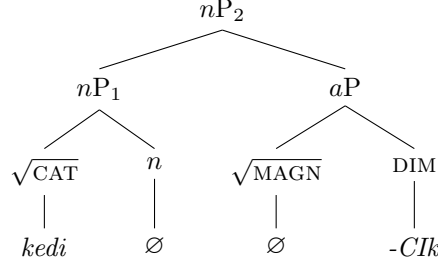
Notice that the first problem is syntactic in nature. The syntactic category of the entire derivation is adjectival, yet *kedicik* is a noun. One could work around this problem by position a null *n* head above the *aP* but it is not obvious why *kedicik* cannot ever be used as an adjective and a null *n* is always required for all the nominal diminutives.

The second problem is semantic in nature. Technically, the derivation in (40) is syntactically valid. $\sqrt{\text{CAT}}$ is a root and DIM is a category-defining functional head and there is nothing that prevents them from combining. However, this combination is both intuitively wrong and also semantically leads to a type mismatch. Intuitively, the diminutive on an adjectival scale denotes a minimal (non-zero) degree associated with the scale dimension. For example, *kısacık* denotes the property associated with a *minimal degree of height*. In contrast, *kedicik* ‘little cat’ does not denote a property associated with a *minimal degree of catness*. A diminutive cat is still a cat but only a very small one. As for the type mismatch, assuming that common nouns are of type $\langle e, t \rangle$ with no inherent ordering among the individuals denoted by the common noun, combining the function DIM directly with the nominal root leads to a type mismatch, i.e. semantic ill-formedness.

Now that we have shown that the diminutive affix on nominal bases does not have the same underlying structure as in adjectival bases, we turn to an alternative which does not suffer from the problems mentioned above. We argue that the syntactic structure of nominal diminutives is simply that of an adjectival modification with a null

adjectival root as in (41). We assume this null adjectival root to be an underspecified magnitude scale ($\sqrt{\text{MAGN}}$) that is compatible with various interpretations such as *size*, *height*, *volume*, etc.

(41) *kedicik* ‘little cat’ (Correct representation)



The structure in (41) does not suffer from the previous type mismatch problem. Notice that *aP* can easily modify the meaning of the *nP*, since it is predicted to denote a function of type $\langle e, t \rangle$ in our semantic system. In (42), we illustrate a sample derivation of (41).

- (42)
- a. $\llbracket nP_1 \rrbracket = \lambda x. x$ is a cat
 - b. $\llbracket aP \rrbracket = \lambda y. \text{magn}(y) \in \text{BR}(\text{magn})$
 - c. $\llbracket nP_2 \rrbracket = \lambda x. (\text{magn}(x) \in \text{BR}(\text{magn})) \wedge (x \text{ is a cat})$

The final meaning ascribed to *kedicik* ‘little cat’ is a property of being cats whose magnitude lies in the *bottom range* of the contextual scale. This correctly captures the intuition that the diminutive form does not minimize the catness, but specifies a low magnitude.²²

²²Before concluding this subsection, we would like to address two comments by the editor, Olivier Bonami, regarding our treatment of nominal diminutives. First, the editor asks whether a sense of bottom rangeness is also involved in nominal diminutives. More specifically, does *kedicik* mean “a cat in the bottom range of size” rather than simply “a somewhat small cat”? We believe that nominal diminutives do indeed involve intensification toward the lower end of the relevant scale. Admittedly, this meaning can be harder to detect with forms like *kedicik*, since diminutives in natural language often develop affective or idiomatic uses (e.g. as terms of endearment for regular-sized cats). These uses may at times obscure the endpoint interpretation, but they do not rule it out. The same point can be made for English: *little cat* can be used endearingly for a lion without altering its core compositional meaning. Moreover, bottom-range evaluation is always contextual. For *kedicik*, the human scale is often the relevant one, in which case cats already count as maximally small. But in other cases the endpoint meaning is fully transparent: *filcik* ‘tiny elephant’ or *evcik* ‘tiny house’ clearly denote entities in the lowest portion of the size scale for elephants or houses. For instance, it would be infelicitous to call a slightly-below-average house an *evcik*, since the form requires reference to the bottom range rather than just “below the norm.”

Second, the editor asks why we do not simply treat adjectival and nominal diminutives as two different items, given that they display both morphophonological differences and apparent semantic differences—much like English plural *-s* and present 3sg *-s*, or agentive *-er* and comparative *-er*. We acknowledge that this is a natural first hypothesis. However, our analysis provides a principled way of unifying the two. The morphophonological differences follow from independent structural properties of adjectival versus nominal environments. As for meaning, we do not take adjectival and nominal diminutives to have distinct semantics. The discussion above illustrates why we posit a common core, namely the “bottom range” contribution of the diminutive morpheme in both domains. The fact that nominal diminutives more frequently give rise to idiosyncratic or conventionalized meanings, whereas adjectival diminutives often look more compositionally transparent, could be related to differences in the syntactic configurations in which they appear, rather than pointing to two entirely separate morphemes. On our account, then, both adjectival and nominal diminutives instantiate the same item, whose semantics is stable but whose realization is shaped by structure.

4.4 Back to inner-domain effects

In the previous sections, we have developed a reductionist account of nominal and adjectival diminutives where the syntactic (featural), semantic (interpretive), and morphological (realizational) properties of the diminutive morpheme are constant and the differences between adjectival and nominal diminutives lie in the syntactic structure. Our analysis so far accounts for a range of facts discussed in Section 3. These are i) [k] deletion in adjectival diminutives but not in nominal diminutives, ii) selectional restrictions on adjectival diminutives but not on nominal diminutives (due to attachment to roots vs. attachment to phrases), and iii) the semantic contribution of the diminutive suffix to adjectival diminutives as a minimizer and nominal diminutives as a size/magnitude diminutive.

There are, however, some outstanding facts that still need an account. These are 1) different stress-shifting behavior observed in nominal and adjectival diminutives, 2) vowel epenthesis in monosyllabic adjectival diminutives, 3) idiosyncratic selection of adjectival roots, and 4) the possibility of special idiomatic interpretations with nominal diminutives. All of these outstanding properties are associated with Level I effects (Kiparsky, 1982), also captured as inner-domain phenomena in the DM literature (Embick, 2010; Embick & Marantz, 2008).

Cyclic phonological effects such as stress-shifting affixes have long been noted in the literature, dating back to the Sound Pattern of English (Chomsky & Halle, 1968). These affixes have been analyzed as Level I affixes by Kiparsky (1982). In the recent Minimalist literature and Distributed Morphology literature, such effects have been associated with the *root phase*, i.e. the first phase that includes the category neutral root. This first-phase has also been called the inner domain (Embick & Marantz, 2008). Many have considered derivational affixes to define cyclic phase heads (Embick, 2010; Embick & Marantz, 2008; Marantz, 1997, 2007). In Chomsky’s (2000; 2001) original formulation of phases, complements of phase heads are sent to the interfaces for interpretation. Marvin (2002) builds on this and Marantz’s proposal that category-defining heads are phases to derive cyclic phonological effects through a phase based syntax. However, Embick (2010) showed that some cyclic phonological effects and inner-domain effects span beyond the first phase head. Thus, he proposed (43) for the cyclic spell-out of syntactic material.

- (43) Cyclic spell-out (Embick, 2010, 18)
When a cyclic head is merged, it triggers the spell-out of cyclic domains in its complement.

The definition in (43) ensures that the root is spelled out along with its category-defining head as well as some other material excluding the next phasal head (e.g., a category changing head). In the following, we assume Embick’s definition of cyclic spell-out to account for the inner-domain effects outlined earlier.

Now that our assumptions regarding the cyclic spell-out of syntactic material is in place, let us review the stress patterns associated with the adjectival and nominal diminutives. As discussed in detail in Section 3, both are stress-shifting. Thus, they are both part of the inner domain. However, adjectival diminutives are pre-pre-stressing

while nominal diminutives display regular word final stress. These are illustrated in Table 19 for ease of reference.

Table 19 Stress patterns across diminutives

Adjective	Turkish	Diminutive
short	kı.sa ^ˈ	kı.sa.çık ^ˈ
soft	yu.mu.şak ^ˈ	yu.mu.ş.a.çık ^ˈ
Noun	Turkish	Diminutive
book	ki.ta ^ˈ p	ki.ta.p.çık ^ˈ
butterfly	ke.le.bek ^ˈ	ke.le.bek.çık ^ˈ

Özçelik (2023) analyzes a range of stress-shifting affixes in Turkish including *-Cık* and argues that *-Cık* has a lexically determined extrametrical structure as in (44). The extrametricality of *-Cık* is due to a lexical requirement that imposes attachment to a binary foot.²³ In addition, Özçelik (2023) argues that this foot structure is trochaic, resulting in the pre-pre-stressing pattern observed in adjectival diminutives as illustrated in (45).

(44) $()_{Ft}.\langle\sigma\rangle$ (Özçelik, 2023, 510)

(45) yu.(mu.ş.a). $\langle$ çık $\rangle$ ‘very soft’ (Özçelik, 2023, 510)

A welcome prediction of Özçelik’s account is that it also accounts for vowel epenthesis in monosyllabic adjectival bases. The lexical requirement on *-Cık* to attach a trochaic binary foot is violated when it combines with a monosyllabic base. This is illustrated in (46).

(46) a. $*(d\acute{a}r).\langle$ çık $\rangle$ ‘very narrow’ (Özçelik, 2023, 510)

b. $(d\acute{a}r.a).\langle$ çık $\rangle$ ‘very narrow’ (Özçelik, 2023, 510)

Özçelik’s (2023) account works perfectly for adjectival diminutives but does not capture nominal diminutives where stress assignment is regular (word final). In the following, we largely adopt Özçelik’s core proposal that the diminutive morpheme has a lexically defined prosodic structure that results in pre-pre-stressing and vowel epenthesis in monosyllabic adjectival diminutives. We further argue that the nominal diminutives simply fall outside the scope of the lexical prosodic structure introduced by the diminutive morpheme, simply because of the locality constraints on morphologically triggered phonological effects.

Building on the vast literature on morphologically conditioned phonology, and the Cophonology Theory in particular (Inkelas, 1997; Inkelas & Zoll, 2005, 2007; Orgun, 1996; Sande, Jenks, & Inkelas, 2020), we assume that the diminutive morpheme is associated with the lexically defined prosodic information proposed by Özçelik (2023)

²³Özçelik (2023) argues that Turkish lacks foot structure except in lexically defined cases. See Özçelik (2023) for a detailed analysis of prosody in Turkish.

in (44). We do however slightly differ from Özçelik in that we do not take the affix *-CIk* as a morpheme. Instead, the relevant prosodic properties are associated with the abstract morpheme DIM. In particular, we follow Sande et al. (2020) who argue that cophonologies are introduced as part of the Vocabulary Items. Lexical content of the diminutive morpheme DIM is given in (47).

- (47) Lexical content of the Diminutive Morpheme
- a. Syntax: $\{a, \prec, \text{br}\}$ (or simply DIM)
 - b. Semantics: $\llbracket \text{DIM} \rrbracket = \lambda g_{\langle e, d \rangle}. \lambda x. g(x) \in \text{BR}(g)$

The Vocabulary Item associated with the diminutive morpheme based on the framework proposed by Sande et al. (2020) is given in (48).²⁴

- (48) Vocabulary Item for the diminutive morpheme
- a. Featural Content: $\text{DIM} \leftrightarrow -CIk$
 - b. Prosodic Content: $(\sigma.\sigma)_{Ft}. \langle CIk \rangle$

The properties of the diminutive morpheme DIM and the associated Vocabulary Item are constant across adjectival and nominal diminutives. The differences in their phonological behavior simply follow from the syntactic structures of the two constructions. Cophonologies have been shown to be constrained in what they can scope over. Rolle (2008) has argued that heads scope over their complements whereas specifiers scope over their heads. Recently, Sande et al. (2020) have argued that cophonologies scope over the phase in which they are phonologically interpreted. A closer look at the syntactic structures of adjectival diminutives and nominal diminutives reveals that adjectival bases are within the scope of the cophonology in (48-b) whereas nominal bases are not.

- (49) *küçücük* ‘very small’ (adjectival diminutive)
- $$\begin{array}{c}
 aP \\
 \swarrow \quad \searrow \\
 \sqrt{\text{SMALL}} \quad \text{DIM} \\
 | \quad \quad | \\
 küçü \quad (\sigma.\sigma)_{Ft}. \langle CIk \rangle
 \end{array}$$

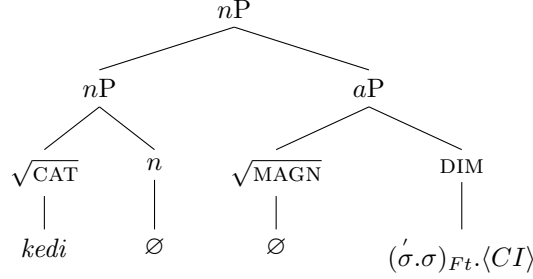
In (49), the cophonology associated with the diminutive morpheme scopes over the scalar root as they are in a local-enough relation. DIM c-commands the root and c-command is sufficient to establish the local relation that is necessary for the cophonology to apply. In fact, McPherson (2014) and McPherson and Heath (2016)

²⁴It should be noted that this is a simplified version of the VI structure proposed by Sande et al. (2020). We have chosen this simplification only for expository purposes. Sande et al. (2020) also encode cophonological content which is a partial constraint ranking that overrides a default ranking. Clearly, we need such a partial ranking to override the default regular word-final stress in Turkish for adjectival diminutives. However, we do not pursue a detailed OT account of the stress-shifting phenomena associated with the diminutive suffix as it goes beyond the scope of this paper. Our main intention in this paper is to show that affixes can be heads or phrasal adjuncts. We leave a detailed phonological analysis of the diminutives for future work.

argued that morpheme-specific phonological rules only scope over elements that are c-commanded by the morpheme.

Compare the syntactic structure in (49) with the one in (50). In (50), the diminutive root is not in a position where it can c-command the nominal root. Thus, the cophonology introduced by the VI associated with the diminutive morpheme cannot scope over the nominal root, bleeding pre-pre-stressing and vowel epenthesis.

(50) *kedicik* ‘little cat’ (nominal diminutive)



Notice, however, that both the nP and aP are in the same cyclic spell-out domain given the assumption in (43), even though n and a are distinct phase heads. They will both be spelled out once the maximal nP is in the complement of a higher phase head (e.g., a determiner). The fact that the cophonology introduced by the diminutive affix is not applicable and yet the nP and aP are spelled out in the same phase as a single prosodic word simply derives the regular word-final stress in nominal diminutives.

In Section 3, we showed that the adjectival bases that combine with the diminutive suffix are quite limited. Although they are all scalar dimensional adjectives, not all such scalar adjectives are combinable with *-Cik*. There is a degree of idiosyncratic selection that restricts the set of scalar roots that can combine with the diminutive suffix (see Table 5). In contrast, nominal adjectives are quite productive. Any noun whose meaning is compatible with diminutivization can combine with *-Cik*. In DM, productivity is usually associated with whether a morpheme is attached directly to a root or a higher phrase. Morphemes that attach to the root are less productive compared to morphemes that attach to larger phrases (Marantz, 2001). This captures the productivity of *-Cik*. In adjectival diminutives, the diminutive morpheme directly attaches to the scalar root and thus it can have idiosyncratic selectional restrictions, which limits its productivity. On the other hand, in nominal diminutives, the diminutive morpheme is not directly attached to the nominal root. Instead, the adjectival phrase containing the diminutive morpheme combines with the nP that contains the nominal root. This predicts less idiosyncrasy and more productivity.

Up till this point, we have provided an account of stress-shift, vowel epenthesis across nominal and adjectival diminutives, and the (semi-)productivity of the diminutive suffix. The only remaining issue is the possibility of idiomatic readings in nominal diminutives. For example, the diminutive nominal in (51) has a transparent reading and an idiomatic reading.²⁵

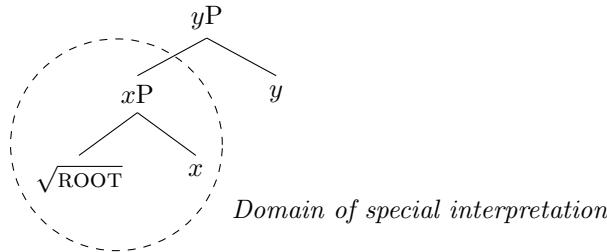
²⁵See Table 9 for more examples.

- (51) maymun-cuk
 monkey-DIM
 ‘little monkey’ (transparent)
 ‘picklock/master key’ (idiomatic)

There is an ostensible difference between adjectival diminutives and nominal diminutives in terms of the availability of idiomatic interpretations. We do not have any instances of idiomatic reading in adjectival diminutives but we do have many such instances in nominal diminutives. The crucial point to note, though, is that idiomatic readings are possible with nominal diminutives but not necessary. In fact, many nominal diminutives do not have any idiomatic readings (e.g., *kedi-cik* ‘diminutive cat’, *arı-cık* ‘little bee’, etc).

The availability of idiomatic readings with affixes has been associated with root attachment (Arad, 2003) and inner domain attachment (Embick, 2010; Embick & Marantz, 2008). Arad (2003) argued that the roots can be assigned a variety of interpretations that are neither predictable nor necessarily related to each other. The interpretation of a root is fixed once it combines with a category-defining head and cannot be changed by any other word formation operations (Arad, 2003).

- (52) Interpretation of roots (Arad, 2003)



Arad’s proposal predicts that xP in (52) can produce special interpretations whereas yP cannot as y does not directly combine with the root.

Although some tension may seem to exist between Arad’s proposal and our analysis of nominal diminutives, illustrated in (21) and (41), an anonymous reviewer rightly notes that the literature often treats special meanings associated with idioms and those arising from polysemy interchangeably. However, Marantz (2013) emphasizes the need to distinguish these two notions. Specifically, Marantz (2013, 105) states:

Marantz (2001) is confusing, if not simply wrong, in conflating the notion of “idiom” with the notion of “special meaning” or “meaning choice” associated with polysemy. For the issue of root (and likely functional morpheme) polysemy, the relevant locality domain for “fixing” meaning appears to be the phase, while for idioms, the domain is clearly larger.

In light of this, Arad (2003)’s proposal should be understood as a restriction on root polysemy. By contrast, the idiomatic meanings associated with nominal diminutives should be classified as idioms, which arise after polysemy resolution and pertain to larger linguistic structures (Marantz, 2013).

As illustrated in (21) and (41), the diminutive suffix *-Cik* forms a complex adjectival phrase and, crucially, does not directly combine with the nominal root. Nonetheless, it can still yield special idiomatic interpretations, which differ fundamentally from root polysemy. Under the assumption that idiomatic expressions are also domain-sensitive (e.g., verbal idioms are constrained within the domain of the external argument, as noted by Marantz (1997)), the nominal root and the diminutive affix (*aP*), though not directly combining, belong to the same spell-out domain. This shared domain enables the assignment of idiomatic interpretations. Therefore, our analysis accounts for the idiomatic meanings of nominal diminutives without requiring modifications to our proposal.

One prediction of our proposal is that special idiomatic interpretation should be available in scenarios where a noun is modified by an adjective that is not an affix as long as they are in the same spell-out domain. This prediction is borne out as illustrated by the examples in (53) – (55).

- (53) kaba-kulak
bulky-ear
'big, bulky, puffy ear' (transparent reading)
'mumps' (special idiomatic reading)
- (54) kırmızı çizgi
red line
'a red line' (transparent reading)
'limit/boundary' (special idiomatic reading)
- (55) beyaz masa
white desk
'a white desk' (transparent reading)
'municipal solutions office' (special idiomatic reading)

In (53) the word *kabakulak* 'mumps' is commonly spelled as a single word and acts as a single prosodic word. Its default interpretation is the idiomatic reading 'mumps'. However, it can be transparently interpreted as 'swollen, puffy, or bulky ear' as well. On the other hand, (54) and (55) are commonly spelled as two words and act as two prosodic words with two distinct primary stresses.²⁶ Thus, we observe that adjectival modification can lead to special idiomatic interpretations regardless of whether the adjective is an affix as in *-Cik*, an adjectival compound as in (53) or as separate prosodic words as in (54) and (55).²⁷ The crucial point though is the position of the adjective. Special idiomatic interpretation is available as long as the noun and the adjective are in a local-enough relation (which we assume to be the same spell-out domain following Embick and Marantz (2008)). Once the locality between the noun and the adjective is broken, idiomatic readings are lost. This is illustrated in (56) – (58) where the indefinite determiner *bir* appears between the adjective and the noun.

²⁶Our descriptions regarding the prosodic word status of these examples are impressionistic and may not be as reliable as an acoustic study.

²⁷Clearly syntactic structures in a single spell-out domain can get to be parsed as a single prosodic word or multiple prosodic words. We have nothing to contribute regarding how prosodic words are determined.

- (56) kaba bir kulak
bulky one ear
'big, bulky, puffy ear' (transparent reading)
*'mumps' (special idiomatic reading)
- (57) kırmızı bir çizgi
red one line
'a red line' (transparent reading)
*'limit/boundary' (special idiomatic reading)
- (58) beyaz bir masa
white one desk
'a white desk' (transparent reading)
*'municipal solutions office' (special idiomatic reading)

While providing a detailed analysis of the indefinite determiner in Turkish is beyond the scope of this paper, we assume that it signals a phase boundary between the adjective and the noun resulting in distinct spell-out domains for the adjective and the noun in (56) – (58).

To summarize this section, our proposal accounts for the stress-shifting and vowel epenthesis properties of both adjectival and nominal diminutives. This account also captures the availability of special idiomatic interpretations in nominal diminutives without any additional stipulations. This concludes the discussion of all the outstanding issues introduced at the outset of this section.

5 Further Considerations

In this section, we discuss two seeming exceptions to the analysis presented in this paper. One concerns the availability of *sıca-cık* 'agreeably hot'. Under the assumption that *sıcak* 'hot' and *soğuk* 'cold' are derived from the same scale, it is surprising how *sıca-cık* is present in the system, while **soğu-cuk* is ungrammatical. In Section 5.1, we are going to entertain two possibilities to address this concern.

The second puzzle concerns the following observation: A number of adjective + *-Cık* combinations appear as exceptions to our selectional restriction regarding the adjectival diminutives discussed in Section 3. These are adjective + *-Cık* combinations that act as place names. Table 20 provides a small list.

Table 20 Seeming Exceptions: Place names consisting of adjective + *-Cık*

English	Turkish	Diminutive	Type of the Adjectival Root
near	yakın	yakın-cık	adjective lower on the scale
cool	serin	serin-cık	adjective lower on the scale
new	yeni	yeni-cık	adjective lower on the scale
long	uzun	uzun-cuk	adjective upper on the scale
deep	derin	derin-cık	adjective upper on the scale
beautiful	güzel	güzel-cık	evaluative adjective
yellow	sarı	sarı-cık	non-gradable adjective
flat	düz	düz-cük	non-gradable adjective

Hence, the question is: If *-CIK* needs to combine with adjectives on the lower end of a scale, how are the combinations listed in Table 20 are possible.

5.1 Agreeable hotness and the scale of *heat*

Throughout the paper, we have treated DIM as a functor that selects the bottom range of a scale and argued that it combines directly with root scales. This explains why the diminutive suffix is only compatible with gradable adjectives whose denotations lie at the lower end of a scale. The only exception we are aware of is the word *sıcacık* ‘pleasantly/agreeably warm’. An anonymous reviewer observes that this word poses a problem for the denotation we provided for the diminutive morpheme. The problem arises most clearly if we assume that *sıcak* ‘hot/warm’ and *soğuk* ‘cold’ are defined on a single HEAT scale, as illustrated in Figure 4 below:

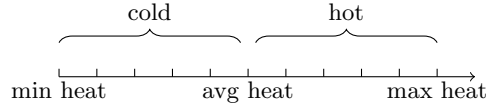


Fig. 4 HEAT scale

Given the definition of the diminutive morpheme as a bottom-range operator, we would expect to find **soğucuk* ‘diminutive cold’ but not *sıca-cık* ‘diminutive hot/warm’. The facts point to the opposite: **soğucuk* is unattested, while *sıca-cık* is fully acceptable. This is the only exception we know of, and we acknowledge the problem. In the following, we discuss ways in which the problem can be handled within the proposed system.

Before presenting the discussion, let us articulate our intuition regarding *sıcacık*. As the reviewer notices, *sıcacık* means ‘pleasantly warm’.²⁸ On the HEAT scale, ‘pleasantly warm’ corresponds to a vague area that is on the warm/hot side of the scale but is just around the bottom range of what counts as warm/hot. We sketch this in Figure 5. The red bar indicates the bottom range of hotness, corresponding to the denotation of *sıcacık*.

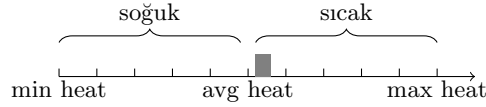


Fig. 5 Bottom range of hotness denoted by *sıcacık* ‘pleasantly warm’

One might try to derive *sıcacık* by first applying UP to the HEAT scale and then applying DIM to the resulting subscale. However, in our system this composition is ill-typed:

²⁸This is another reason why we disagree with Türk (2020), who treated *-Cık* as an augmentative. An augmentative reading here would denote an augmented heat, which is not what *sıcacık* conveys.

- (59) a. $\llbracket \text{UP} \rrbracket : \langle \langle e, d \rangle, \langle e, t \rangle \rangle$
 b. $\llbracket \text{DIM} \rrbracket : \langle \langle e, d \rangle, \langle e, t \rangle \rangle$

Accordingly, $\text{UP}(g)$ returns a predicate of type $\langle e, t \rangle$, which cannot serve as the $\langle e, d \rangle$ argument that DIM requires. Even if this were possible, it would open up the possibility of applying DIM on top of UP elsewhere, practically enabling combination with upper end adjectives.

One way to overcome the type mismatch would be to introduce a separate degree-shifting variant, $\text{UP}^{\text{deg}} : \langle \langle e, d \rangle, \langle e, d \rangle \rangle$, that returns a measure function restricted to values $\geq s(g)$. However, this would not necessarily overcome the second issue, that is the overgeneration problem.

On an alternative view, one could argue that the scale associated with heat in *sıcacık* is a scale of HOTNESS as in Figure 7, distinct from the scale of COLDNESS in Figure 6. The coldness scale involves minimal and maximal coldness, while the hotness scale involves minimal and maximal hotness. Under such assumptions, the diminutive works as usual, picking out the bottom range of hotness.

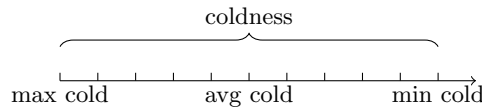


Fig. 6 COLDNESS scale

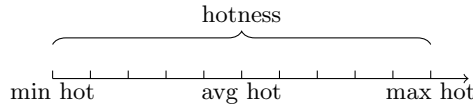


Fig. 7 HOTNESS scale

Notice that this view does not preclude the existence of a single heat scale along with these additional scales. Rather, it amounts to the conflation of the heat and the degree shifting versions of the up/down functors at the lexical root level for some adjectives. That is, it is possible that some roots denote measure functions that are only defined for entities which are already below (for coldness scale) or above (for hotness scale) the standard degree. Setting aside potential redundancy, we believe that this view is more in line with our analysis of *-CIK* in Turkish, given that it does not overgenerate other forms while also accounting for the meaning associated with *sıcacık* ‘pleasantly/agreeably hot’. Positing such lexical roots representing different scales, namely *coldness* and *hotness*, can also explain why **soğucuk* is ungrammatical: among the two heat-related scales, only the hotness scale is in the domain of the function denoted by *-CIK*, hence the ungrammaticality of **soğucuk*.

We cannot currently comment on the metaphysical foundations of having two related HEAT scales in our linguistic ontology. We take this as a representational device

in the grammar, not a claim about the metaphysics of temperature. However, distinguishing two related scales finds some cross-linguistic support. It is well known that certain antonym pairs are not symmetrical: for example, questions about age are usually framed with *old*, while *young* is marginal without special contextual support (see for example Kennedy 2007; Ljung 1974). Indeed, in a context where Bruce is known to be young, (60-a) is felicitous, but in a context where Bruce is known to be old, (60-b) is odd.

- (60) a. How old is Bruce?
b. #How young is Bruce?

By contrast, for the hot–cold pair we do not see a clear neutral adjective. In a context where the room is known to be cold, it is infelicitous to ask (61-a); in a context where the room is known to be hot, it is infelicitous to ask (61-b). Unlike the case with *old/young*, neither adjective seems more marginal in (61).

- (61) a. How hot is the room?
b. How cold is the room?

The neutral question here would be *What is the temperature?*. These contrasts may be evidence that in addition to a core HEAT scale, natural language may recruit scales of HOTNESS and COLDNESS.

5.2 Place names

In Table 20, we presented all sorts of seeming exceptions to our analysis of *-Cik*. For convenience, we repeat those exceptions in Table 21 again.

Table 21 Seeming Exceptions: Place names consisting of adjective + *-Cik*

English	Turkish	Diminutive	Type of the Adjectival Root
near	yakın	yakın-cık	adjective lower on the scale
cool	serin	serin-cik	adjective lower on the scale
new	yeni	yeni-cik	adjective lower on the scale
long	uzun	uzun-cuk	adjective upper on the scale
deep	derin	derin-cik	adjective upper on the scale
beautiful	güzel	güzel-cik	evaluative adjective
yellow	sarı	sarı-cık	non-gradable adjective
flat	düz	düz-cük	non-gradable adjective

Adjectives on the lower end of a scale that normally do not combine with *-Cik* (as shown on Table 5) are acceptable as place names. Adjectives on the upper end of a scale are completely fine as place names. Likewise, evaluative adjectives and even non-gradable adjectives are acceptable when the diminutive form is a place name.²⁹ The crucial thing to note is that the seeming exceptions are all nouns (place names

²⁹Some of these seemingly exceptional diminutives are also used as surnames.

or surnames) whereas adjectival diminutives are adjectives. This is reflected on their syntactic and morphophonological properties.

For expository purposes, let us call the two types of diminutives as in (62).

- (62) a. Adjectival diminutive = [adjective + -*CIk*]_a
b. Place diminutive = [adjective + -*CIk*]_n

The first difference between adjectival diminutives and place diminutives is their syntactic category. Adjectival diminutives are adjectives. They combine with other noun phrases as adjectival modifiers as shown in (63).

- (63) a. küçük karınca
small ant
‘the small ant’
b. küçü-cük karınca
small-DIM ant
‘the very small ant’

On the other hand, place diminutives behave like nouns and form noun-noun compounds with the noun they modify as evidenced by the possessive suffix on the head noun. This is illustrated in (64).

- (64) a. güzel köy
beautiful village
‘beautiful village’
b. güzel-cik köy-ü
beautiful-DIM village-POSS
‘Güzeltik village’
c. *güzel-cik köy
beautiful-DIM village
‘beautiful village’

The second difference between adjectival diminutives and place diminutives regards the stress assignment. In adjectival diminutives, the diminutive suffix is pre-pre-stressing. In contrast, the diminutive suffix does not have any impact on the stress assignment in place diminutives, leaving the stress on the penultimate syllable as is the case on most place names in Turkish (Sezer, 1981). Table 22 illustrates the stress patterns.

Table 22 Stress patterns in -*CIk* diminutives.

Adjectival Diminutive			Place Diminutive		
English	Turkish	Diminutive	English	Turkish	Diminutive
short	kı.sa ^ˈ	kı.sa-cık ^ˈ	beautiful	gü.zel ^ˈ	gü.el-cık ^ˈ
thin	in.ce ^ˈ	in.ce-cık ^ˈ	thick	ka.lın ^ˈ	ka.lın-cık ^ˈ
small	kü.çük ^ˈ	kü.çük-cük ^ˈ	deep	de.rin ^ˈ	de.rin-cık ^ˈ
tiny	mi.nik ^ˈ	mi.ni-cık ^ˈ	long	u.zun ^ˈ	u.zun-cuk ^ˈ

A third difference between adjectival diminutives and place diminutives appears in monosyllabic adjectives. When *-Cik* combines with a monosyllabic adjectival base to create an adjectival diminutive, an epenthetic vowel shows up between the root and the diminutive suffix. With place diminutives epenthesis does not occur. This is shown in Table 23.

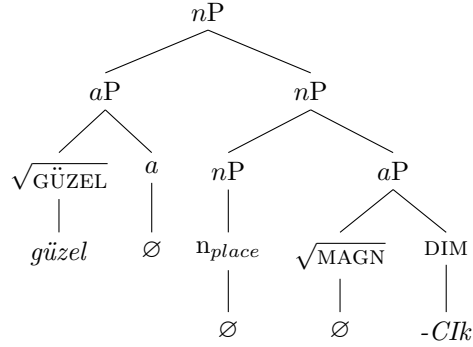
Table 23 Vowel epenthesis in adjectival diminutives

Adjectival Diminutive			Exceptional Diminutive		
English	Turkish	Diminutive	English	Turkish	Diminutive
narrow	dar	dar-a-cık	flat	düz	düz-cük
young	genç	genc-e-cik	one	bir	bir-cik ¹
one	bir	bir-i-cik			
little	az	az-ı-cık			

¹Monosyllabic adjective roots are rare and their diminutivized forms are even rarer. This example (*bircik*) is a surname whereas the one above (*düzcük*) is a place name.

In all the relevant respects, place diminutives behave like nominal diminutives. The only ostensible exception is stress assignment. However, in Turkish, place names have exceptional stress (Sezer, 1981). Thus, we hypothesize that place diminutives are simply constructions where the head noun is null with a feature indicating place (or name). The structure of place diminutives is like in (65).³⁰

(65) *Güzelcik* ('Beautifultown')



6 Conclusion

In this paper, we have analyzed the syntactic, semantic and morphophonological properties of the diminutive suffix *-Cik* in Turkish. We have argued that *-Cik* is a category-defining *a* head with the diminutive semantics. We have shown that *-Cik*

³⁰It is unclear to us whether the concept of placeness in a name is encoded in the nominalizing head *n* or in a classifier head above *n*. Either way should work for our purposes.

can create both adjectival and nominal diminutives. Both adjectival diminutives and nominal diminutives display some inner-domain effects such as stress-shifting, vowel epenthesis and special idiomatic readings. Yet, adjectival diminutives differ from nominal diminutives in some non-trivial ways. To account for these facts, we proposed an account of diminutives in Turkish where the adjectival diminutives are formed by attaching the diminutive morpheme directly to a scale root whereas nominal diminutives are formed by first establishing a diminutivized adjectival phrase and then attaching it to a nominalized root as a modifier.

A central claim of this paper has been that gradable adjectives are syntactically built from abstract scales. This approach captures the inherent connection between positive and negative members of a scalar pair, such as big and small, by deriving them from a shared scale. One result of such a claim has been the availability of lower end of scales for diminutivization in Turkish. Our analysis thus converges with proposals such as Fábregas and Marín (2018) and Vanden Wyngaerd, Caha, and De Clercq (2025), which posit a rich internal structure for gradable adjectives.

A broader conclusion of our analysis is that derivational affixes can be more complex than their surface form suggests, associating with their bases either as heads or as modifiers. Our findings support earlier claims about multiple modes of affix attachment, particularly in the case of diminutives (Gouskova & Bobaljik, 2022; Wiltschko & Steriopolo, 2007). What is striking in our case is that the same surface affix is analyzed as the realization of two distinct morpho-syntactic structures: its internal structure and mode of attachment vary depending on whether it combines with a nominal or an adjectival base. An open question, which we leave for future research, is whether such variability in attachment size and mode is unique to diminutive morphology or extends to other affixes in Turkish and beyond.

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